LAURA OZORIA

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EDUCATION

Bucknell University Lewisburg, PA, Expected May 2027
Majors: Computer Science and Engineering, BS; Film and Media Studies, BA
Relevant courses: Computer Systems; Algorithm Design and Analysis; Data Structures and Algorithms; Software Engineering and Design; Machine Learning; Discrete Structures; Statistics for Engineers; Film Production.

SKILLS

Programming Languages: Python, C++, Java, C, SQL, JavaScript, R, HTML, CSS, RISC-V Assembly, Rust
Frameworks: Flask, Spring Boot, Next.js, Node.js, Bootstrap, PyTorch, TensorFlow, JavaFX, Junit, React
Tools: Figma, Unity, Godot, Onshape CAD, Git, Linux, Bash, CI/CD workflows, AWS
Languages: Spanish (Native), Japanese (Intermediate)

EXPERIENCE

Bucknell University Lewisburg, PA
Software Programmer for Color Analysis Program August 2024 – Present
- Develop a web-based version of KALMUS film analysis software using HTML, Flask, and Bootstrap, expanding accessibility to 40+ student researchers.
- Expand a Python film color analysis toolkit (KALMUS) by developing 5+ new features used in the web adaptation of the package.
- Improve color analysis accuracy by 10% through the integration and tuning of machine learning models.
- Designed a SQLite-backed metadata system integrated with the IMDb API to enable real-time film data retrieval.

Data Science Student Fellow January 2026 - Present
- Collaborate with stakeholders and data science mentors on a semester-long applied data science project.
- Extract and process audio data from video content using Python, Excel, and Premiere Pro to enable quantitative analysis for client stakeholders.
- Support data science outreach initiatives by promoting student engagement in workshops and seminars across campus.

Engineering Peer Mentor August 2024 - Present
- Guided 5 students through resume development and LinkedIn profile optimization to strengthen their professional branding and career preparation.
- Maintained consistent engagement through weekly meetings throughout the semester, with additional availability for on-demand academic support in physics and computer science.

Virtual Reality Research Assistant June - July 2023
- Designed an interactive colorimetry learning program in Unity to teach primary/secondary color theory and color mixing principles through hands-on VR experiences.
- Created Unity-based VR learning tool for Meta Quest 2 platform, enabling standalone access to interactive color theory education.

PROJECTS

Movie Night Scheduler (Hackathon Project) July 2025
Java, Spring Boot, Thymeleaf*, HTML/CSS, JavaScript, Figma, Google Calendar API*
- Developed a proof-of-concept web application **for** group availability matching and shared scheduling logic.
- Designed user flows and UI mockups in Figma and built a responsive frontend using HTML, CSS, and JavaScript.
- Implemented RESTful APIs and integrated Google Calendar API, enabling real-time event synchronization.

Stardew Valley Simulation - Object-Oriented Game (Java, JavaFX) April 2025
- Developed a 2D farming simulation featuring 5+ gameplay mechanics and 10+ managed game objects.
- Applied object-oriented design principles and UML modeling to create a modular, maintainable architecture.
- Completed 5 Agile sprints using test-driven development with JUnit and Scrum methodologies.

Animal Species Identifier - Neural Network (Python, TensorFlow) April 2025
- Engineered and trained in a CNN-based classifier for animal species identification, achieving 80% accuracy on a limited dataset.
- Assessed performance with confusion matrices, accuracy metrics, and dataset distribution analysis.